

untangle::async

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Chapter 1

async

c++ mechanism to run queued callables in an asynchronous fashion

The interface consists of one header file that exposes a generic execution class. It is a wrapper of a queue of callables that are invoked on other thread. The callables are pumped into the queue by invoking an action from the caller thread.

The implementation is using the **actuator** callable, and therefore it exemplifies how a generic callable may improve the code structure, by easily creating interfaces inside a class instead of using external ones.

There are provided two execution modes: one-off and continuous.

requirements

C++23. The interface reports warnings with `std::println`, so it needs a standard library that provides `<print>` — GCC 14 or newer; older `libc++` and `MSVC` releases may not have it yet.

one-off execution



example: one-off execution of a plain function

```
#include <async.hpp>

void f()
{
    std::cout << "f thread " << std::this_thread::get_id() << std::endl;
}

int main()
{
    std::cout << "main thread " << std::this_thread::get_id() << std::endl;

    // declare an execution specialized for an action type; it must be held by a shared pointer,
    // because an async binding keeps only weak ownership of it
    // the name is optional, and identifies the execution in the warnings it reports
    using void_execution = untangle::async::execution<std::function<void(void)>>;
    auto execution = void_execution::create_instance("oneoff");
```

```

std::function<void(void)> action;
// create async binding between the action and f; the execution is passed as a shared pointer,
// which the binding holds weakly - that is what lets the action outlive it safely
void_execution::bind_action_and_function(action, f, execution);
// whenever the action is invoked, a new callable wrapping f will be added to the execution's action queue
action();
// run the execution
execution->run();

// add the execution object to a poll of one's own; the poll answers for what was added to it
untangle::async::execution_poll poll;
poll.add(*execution);

// wait the polled executions to finish
while(poll.is_running())
{
    std::this_thread::sleep_for(std::chrono::milliseconds(1));
}

return 0;
}

```

example: one-off execution of a class method

```

#include <async.hpp>

class A
{
public:
    void f(int x)
    {
        std::cout << "A::f " << x << " thread " << std::this_thread::get_id() << std::endl;
    }
    std::function<void(int)> action;
};

int main()
{
    std::cout << "main thread " << std::this_thread::get_id() << std::endl;

    auto a = std::make_shared<A>();

    // declare an execution specialized for an action type; it must be held by a shared pointer
    using int_execution = untangle::async::execution<std::function<void(int)>>;
    auto execution = int_execution::create_instance("oneoff_method");
    // note both the bound object and the execution must be shared pointers
    int_execution::bind_action_and_method(a->action, a, &A::f, execution);
    // whenever the action is invoked, a new callable wrapping f will be added to the execution's action queue
    a->action(10);
    // run the execution
    execution->run();

    // add the execution object to a poll of one's own; the poll answers for what was added to it
    untangle::async::execution_poll poll;
    poll.add(*execution);

    // wait the polled executions to finish
    while(poll.is_running())
    {
        std::this_thread::sleep_for(std::chrono::milliseconds(1));
    }

    return 0;
}

```

continuous execution



example: continuous execution of a plain function

```

#include <async.hpp>

void f()
{

```

```

    std::cout << "f thread " << std::this_thread::get_id() << std::endl;
}

int main()
{
    std::cout << "main thread " << std::this_thread::get_id() << std::endl;

    // declare an execution specialized for an action type; it must be held by a shared pointer
    using void_execution = untangle::async::execution<std::function<void(void)>>;
    auto execution = void_execution::create_instance("continuous");
    std::function<void(void)> action;
    // create async binding between the action and f
    void_execution::bind_action_and_function(action, f, execution);

    // run the execution
    execution->start();
    // add f to the execution's queue (three times)
    action();action();action();
    // stop the execution
    execution->stop();

    // add the execution object to a poll of one's own; the poll answers for what was added to it
    untangle::async::execution_poll poll;
    poll.add(*execution);

    // wait the polled executions to finish
    while(poll.is_running())
    {
        std::this_thread::sleep_for(std::chrono::milliseconds(1));
    }

    return 0;
}

```

queueing and lifetime

Queueing is thread safe. An action may be invoked from any thread; the execution's list is guarded, and the worker is woken rather than polled for.

stop() ends the execution's working life. The worker runs whatever was queued before the call and then leaves. An action invoked after stop() is refused rather than queued, because it would never run, and the refusal is reported:

```
warning: execution 'continuous' is stopped, action not added
```

A task is refused after stop() for the same reason, and says so in its own words:

```
warning: execution 'continuous' is stopped, task not added
```

A task is also refused before it is ever queued if it could not do its job — no callback to notify, or nothing to run. That one is the caller's own mistake and is caught while they are still on the stack, rather than surfacing later as a failure of a task whose action had in fact succeeded:

```
warning: execution 'continuous' refused a task that cannot report
```

An action bound to an object that has since been destroyed is dropped the same way, rather than ending the process:

```
warning: execution 'continuous' dropped an invalid action: bind: invalid object
```

The worker is detached and cannot be joined. Waiting for an execution therefore means waiting on its running state, which is what execution_poll reports and what the examples above do. The destructor waits on the same state, so an execution that goes out of scope while its worker is still running blocks until the worker is finished rather than leaving it reading freed memory.

The poll holds a pointer to each execution added to it, so an execution withdraws itself in its destructor - from every poll it was added to, since it may be in more than one. A poll destroyed first likewise releases what it still holds, so a poll may be a local. execution_poll::remove() is available for withdrawing one earlier; calling it is not required.

tasks: work that reports back

Everything above queues actions, through `add_action()`. An action is fire and forget: it runs, what it returns is dropped, and the caller hears nothing more about it. A task, through `add_task()`, carries the callback it must notify:

```
auto exec = untangle::async::execution<std::function<int(int)>>::create_instance("scale");
exec->start();

// the callback comes last, after the arguments the action is bound to
exec->add_task([](int n) { return n * 2; }, 21, [](int result) { /* result == 42 */ });
```

A task whose action returns void still reports that it finished, with a callback taking nothing — finished is the message and the result is optional:

```
exec->add_task([](int by) { /* ... */ }, 5, [] { /* finished */ });
```

The callback is the last argument, and the signature cannot say so. A parameter pack cannot be followed by a deducible parameter, so it arrives inside the arguments and `untangle::bind_task()` splits it off. It must be callable with the action's result and return nothing — or callable with nothing at all, when the action returns void. A missing or unusable one is a compile error rather than silence.

Why a task carries its own callback rather than being handed one. The actuator has a callback convention already: pass a trailing callable to an actuator's call operator and it is invoked with each action's result. That convention reads the trailing argument of an invocation, and a queued call has no invocation left to read from — so the callback has to travel with the work. That is what a task is.

`add_task()` answers bool, and the answer is the whole report. It refuses a task the execution can never run — because it is stopped — or one that could never do its job: no callback to notify, or nothing to run. A refused task is dropped, so it never runs and never notifies, and a caller waiting on the callback alone would wait for ever.

Finished does not mean failed. A task whose action throws is not notified: there is no result to hand over, and for a void task no completion to report either. What it threw goes to `on_error`, as a failing action's does. And because a callback runs inside the same try as the task that owns it, `on_error` can fire for a task whose action in fact succeeded — the work was done and only the telling failed.

`on_finished` is not a task's callback. It is raised once per pass, when the queue has drained, and a pass that ran only tasks raises it like any other. A task's own callback has already run by then, inside the same drain and on the same thread. Two notifications, two meanings: one per task, one per drained queue.

Both kinds share one queue and one worker. Within a single pass every action runs before any task, so the two do not interleave in the order they were queued; across passes the queue is still first in, first out.

building the tests

```
cmake -S test -B test/build
cmake --build test/build
ctest --test-dir test/build
```

`test/CMakeLists.txt` fetches googletest at configure time, so the first configure needs network access. The actuator submodule has to be present: `git submodule update --init`.

sanitizers

ASYNC_SANITIZE builds the tests under a sanitizer. It is off by default, because a sanitized build is several times slower and ThreadSanitizer does not ship for every toolchain.

```
cmake -S test -B test/build-asan -DASYNC_SANITIZE=address
cmake --build test/build-asan
ctest --test-dir test/build-asan
```

Accepted values are address, thread, undefined, or empty. Anything else is refused at configure time rather than passed through to the compiler.

Use a separate build directory per sanitizer. address and thread instrument the same accesses in incompatible ways and cannot be combined, so one build directory is one sanitizer.

The two answer different questions. ThreadSanitizer names both sides of a data race, which is how the queue's races were diagnosed. AddressSanitizer reports a use-after-free, which is how the dangling attachment was. Neither sees everything: a defect reached through a destroyed `std::mutex` throws `std::system_error` from inside `libsystem` before any instrumented load runs, so both sanitizers stay silent. Reach for a direct instrumented read when that happens.

Chapter 2

Topic Index

2.1 Topics

Here is a list of all topics with brief descriptions:

namespace untangle: functions [15](#)

Chapter 3

Namespace Index

3.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

untangle	Interface to untangle::async::execution class	17
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Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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One execution's place in the attachment graph: who attached it, if anyone	19
untangle::async::execution< actionT >	
Asynchronous template execution class	19
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Poll that reports whether any execution added to it is still running	31
untangle::async::invalid_attachment	
Raised by execution::attach() when the attachment asked for is refused	33

Chapter 5

File Index

5.1 File List

Here is a list of all documented files with brief descriptions:

async.hpp		35
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Chapter 6

Topic Documentation

6.1 namespace untangle: functions

Chapter 7

Namespace Documentation

7.1 untangle Namespace Reference

Interface to [untangle::async::execution](#) class.

7.1.1 Detailed Description

Interface to [untangle::async::execution](#) class.

Chapter 8

Class Documentation

8.1 `untangle::async::attachment` Struct Reference

One execution's place in the attachment graph: who attached it, if anyone.

```
#include <async.hpp>
```

Public Attributes

- `std::weak_ptr< attachment >` `attacher`
The link of the execution that attached this one; empty when nothing has.

8.1.1 Detailed Description

One execution's place in the attachment graph: who attached it, if anyone.

At namespace scope, so it is one type rather than one per specialisation: an execution may be attached by an execution of any specialisation, and the chain is walked without knowing any link's action type. It dies with its execution, so it doubles as a liveness token - a link that still locks belongs to an execution that is still there.

The documentation for this struct was generated from the following file:

- `async.hpp`

8.2 `untangle::async::execution< actionT >` Class Template Reference

Asynchronous template execution class.

```
#include <async.hpp>
```

Public Member Functions

- [execution](#) ()
Constructs an execution with the default name. Call [create_instance\(\)](#) instead.
- [execution](#) (std::string exec_name)
Constructs a named execution. Use [create_instance\(\)](#) for one that is to be bound.
- [~execution](#) ()
Destroys the execution, once its worker has left.
- bool [is_running](#) () const
Is a worker of this execution's still there?
- bool [is_busy](#) () const
Is this execution working through its action queue?
- std::size_t [pending](#) () const
How many actions and tasks are queued and not yet taken by the worker.
- void [run](#) ()
Runs what is queued on a worker of its own, once, and keeps the results.
- void [start](#) ()
Starts a continuous worker that runs whatever is queued until [stop\(\)](#).
- void [stop](#) ()
Stops the worker, after the actions already queued have run.
- template<typename... Args>
bool [add_action](#) (actionT action, Args &&... args)
Queues an action, bound to args, and says whether it was taken.
- template<typename... Args>
bool [add_task](#) (const actionT &action, Args &&... args)
Queues a task, and says whether it was taken.
- template<typename otherT>
bool [add_task](#) (actionT &&action, Args &&... args)
Queues a task as the overload above does, moving the action into it rather than copying it. A temporary and a lambda land here.
- template<typename otherT>
void [attach](#) (const std::shared_ptr< otherT > &other)
Attaches another execution, so that this one triggers it.
- template<typename otherT>
bool [detach](#) (otherT &other)
Detaches an execution previously attached by [attach\(\)](#).
- auto [results](#) () const
The return values of the actions run by [run\(\)](#), in the order they ran.

Static Public Member Functions

- template<typename... Args>
static std::shared_ptr< execution > [create_instance](#) (Args &&... args)
Creates an execution owned by a std::shared_ptr, which is what binding to it requires.
- template<typename classT, typename T>
static void [bind_action_and_method](#) (actionT &action, const std::shared_ptr< classT > &obj, T classT::*method, const std::shared_ptr< execution > &async_exec)
Binds asynchronously an external action to a class function member.
- template<typename T>
static void [bind_action_and_function](#) (actionT &action, T Fn, const std::shared_ptr< execution > &async_exec)
Binds asynchronously an external action to a plain function.

Public Attributes

- `std::function< void(void)>` [action_execute](#)
Connection point that runs this execution's queued actions. Bound to `execute_actions()`.
- `std::function< void(void)>` [action_stop](#)
Connection point that stops this execution. Bound to `stop()`.
- `std::function< bool(void)>` [action_is_running](#)
Connection point that reports whether this execution is running. Bound to `is_running()`.
- `std::function< void(void)>` [on_finished](#)
Called on the worker's thread once the queue has drained. Assigned by the caller.
- `std::function< void(std::exception_ptr)>` [on_error](#)
Called on the worker's thread with whatever an action threw. Assigned by the caller.

Static Public Attributes

- `static constexpr auto default_name = "default_name"`
Name of an execution built unnamed.

8.2.1 Detailed Description

```
template<typename actionT>
class untangle::async::execution< actionT >
```

Asynchronous template execution class.

An async execution provides a mechanism to queue work and execute it sequentially on a separate thread. It may also attach another [execution](#) object and trigger its actions. This way actions of different types may be executed on the same thread.

Two kinds of work may be queued. An action, through [add_action\(\)](#), is fire and forget: it runs, and the caller hears nothing more about it. A task, through [add_task\(\)](#), carries the callback it must notify, and reports what it produced - or, for an action returning void, merely that it finished. A queued call has no invocation left for the actuator's own callback convention to read a callback from, which is why a task carries its own.

The mechanism relies on an "asynchronous binding" created by [bind_action_and_method\(\)](#) or [bind_action_and_function\(\)](#).

Template Parameters

<code>actionT</code>	It represents the type of the action. It is specified as <code>std::function<...></code> and should match the signature of the bound function or class method.
----------------------	--

8.2.2 Constructor & Destructor Documentation

8.2.2.1 `execution()` [1/2]

```
template<typename actionT>
untangle::async::execution< actionT >::execution () [inline]
```

Constructs an execution with the default name. Call [create_instance\(\)](#) instead.

The name identifies this execution in the warnings it reports, and is otherwise unused.

8.2.2.2 execution() [2/2]

```
template<typename actionT>
untangle::async::execution< actionT >::execution (
    std::string exec_name)    [inline], [explicit]
```

Constructs a named execution. Use [create_instance\(\)](#) for one that is to be bound.

Parameters

exec_name	- A name for this execution.
-----------	------------------------------

8.2.2.3 ~execution()

```
template<typename actionT>
untangle::async::execution< actionT >::~~execution ()    [inline]
```

Destroys the execution, once its worker has left.

The worker is detached and cannot be joined, so this waits on the "running" state instead, and returns at once for one that was never started. An attacher holds this execution weakly, so there is nothing to take back out of it: its next pass finds this gone and drops it.

Attention

Safe only because `running__` is cleared as the very last thing the worker does: this object may be freed the moment it reads false, so nothing may follow it in `execute()` or `loop()`.

Remarks

No rule of five, deliberately. An execution is declared once and used in place; the members already delete the copy operations several times over.

8.2.3 Member Function Documentation

8.2.3.1 create_instance()

```
template<typename actionT>
template<typename... Args>
std::shared_ptr< execution > untangle::async::execution< actionT >::create_instance (
    Args &&... args)    [inline], [static]
```

Creates an execution owned by a `std::shared_ptr`, which is what binding to it requires.

An execution built any other way is still usable - [add_action\(\)](#) takes actions directly - but cannot be bound, and the attempt does not compile.

Parameters

args	- Constructor arguments: a name, or nothing for the default name.
------	---

Returns

- A std::shared_ptr owning the new execution.

8.2.3.2 bind_action_and_method()

```
template<typename actionT>
template<typename classT, typename T>
void untangle::async::execution< actionT >::bind_action_and_method (
    actionT & action,
    const std::shared_ptr< classT > & obj,
    T classT::* method,
    const std::shared_ptr< execution< actionT > > & async_exec) [inline], [static]
```

Binds asynchronously an external action to a class function member.

It creates an action as a binding to a class method (by untangle::bind()), and assigns to action a callable that queues it. The execution is held weakly, so the action may outlive it; the action itself is shared with every call queued from it, rather than copied per call.

Attention

Invoking action after the execution has been destroyed throws untangle::invalid_action. The worker catches it and drops the action with a warning.

Parameters

action	[in,out] - An action of type std::function<...>.
obj	- A std::shared_ptr that wraps the bound class object.
method	- Pointer to function member. It is specified as &<class type>::<function member>.
async_exec	- A std::shared_ptr owning the execution the action is queued on. Requiring it here is what keeps an execution that no shared_ptr owns from being bound at all.

8.2.3.3 bind_action_and_function()

```
template<typename actionT>
template<typename T>
void untangle::async::execution< actionT >::bind_action_and_function (
    actionT & action,
    T Fn,
    const std::shared_ptr< execution< actionT > > & async_exec) [inline], [static]
```

~~Binds asynchronously an external action to a plain function.~~

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It wraps Fn in an action and assigns to action a callable that queues it. The execution is held weakly, as in [bind_action_and_method\(\)](#), and the action is shared with every call queued from it rather than copied per call.

Parameters

action	[in,out] - An action of type <code>std::function<...></code> .
Fn	- A plain function, or any callable; moved into the binding.
async_exec	- A <code>std::shared_ptr</code> owning the execution the action is queued on.

8.2.3.4 `is_running()`

```
template<typename actionT>
bool untangle::async::execution< actionT >::is_running () const    [inline]
```

Is a worker of this execution's still there?

True from [run\(\)](#) or [start\(\)](#) until the worker has left, which is the same instant [~execution\(\)](#) waits for and the same one [run\(\)](#) waits for before starting another.

Remarks

[on_finished](#) is raised from inside the drain and reads this as true - the worker is standing in the callback with the rest of its path still to go.

Returns

true - a worker is still there.

8.2.3.5 `is_busy()`

```
template<typename actionT>
bool untangle::async::execution< actionT >::is_busy () const    [inline]
```

Is this execution working through its action queue?

True from the moment an action is added until the last one has returned. Unlike [is_running\(\)](#), which is about the worker thread, this answers "has it room for more work"

- a worker started by [start\(\)](#) is running for its whole life whether busy or idle.

Remarks

Answers for this execution's own actions, not for those it triggers through [attach\(\)](#).

A false answer is durable only for the caller that is itself adding the actions; a true one may stop being true the moment after.

Returns

true - actions are queued, or one is running.

8.2.3.6 pending()

```
template<typename actionT>
std::size_t untangle::async::execution< actionT >::pending () const [inline]
```

How many actions and tasks are queued and not yet taken by the worker.

What the worker has taken into the batch it is running is not pending: a worker running its last action reports 0 here and true from `is_busy()`.

Remarks

Advisory, like `is_busy()`: true the moment it is read.

8.2.3.7 run()

```
template<typename actionT>
void untangle::async::execution< actionT >::run () [inline]
```

Runs what is queued on a worker of its own, once, and keeps the results.

The worker is detached: wait for it with `is_running()` or `execution_poll`. Actions queued while it is draining are run by it too. `results()` is filled by this path only.

Blocks until a worker still resident from a previous run has left, so one execution never has two. `is_running()` reads false at exactly that moment, so a caller that waits for it and then runs again does not wait here at all.

Attention

Never call this from inside an action: the worker that would have to leave is the one making the call, so it waits for itself and hangs.

8.2.3.8 start()

```
template<typename actionT>
void untangle::async::execution< actionT >::start () [inline]
```

Starts a continuous worker that runs whatever is queued until `stop()`.

Unlike `run()`, the worker stays for the life of the execution, so `is_running()` is true whether or not there is work; ask `is_busy()` instead. Results are not collected.

Blocks until a worker still resident has left, as `run()` does.

Attention

Never call this from inside an action, for the reason given on `run()`.

8.2.3.9 stop()

```
template<typename actionT>
void untangle::async::execution< actionT >::stop () [inline]
```

Stops the worker, after the actions already queued have run.

Remarks

An action passed to [add_action\(\)](#) after this call is refused, not queued: [stop\(\)](#) is the end of this execution's working life, and an action accepted then would never run.

8.2.3.10 add_action()

```
template<typename actionT>
template<typename... Args>
bool untangle::async::execution< actionT >::add_action (
    actionT action,
    Args &&... args) [inline]
```

Queues an action, bound to args, and says whether it was taken.

It runs later, on whichever thread drives this execution - its own worker, or an attacher's. Queueing before [run\(\)](#) or [start\(\)](#) is normal.

Attention

A stopped execution refuses and drops the action, silently for a caller that ignores the answer. Nothing is thrown: `untangle::invalid_action` means a dead binding, while a refused action is sound and merely too late.

Remarks

A caller arriving through [bind_action_and_method\(\)](#) or [bind_action_and_function\(\)](#) never sees the answer: those lambdas return `actionT::result_type`, which has no room for it.

It does not notify anyone. The answer below is the last the caller hears: what the action returns is run and dropped, and nothing reports that it finished. Use [add_task\(\)](#) for work that has to report. A trailing callable passed here is an ordinary argument of the action - the actuator's callback convention reads a trailing argument of an invocation, and a queued call has none.

Template Parameters

Args	- The argument types the action is bound to.
------	--

Parameters

action	- The action to queue.
args	- The arguments to bind to action.

Returns

- true - queued, and this execution's worker will run it.
- false - the execution is stopped and the action was dropped.

8.2.3.11 add_task()

```
template<typename actionT>
template<typename... Args>
bool untangle::async::execution< actionT >::add_task (
    const actionT & action,
    Args &&... args) [inline]
```

Queues a task, and says whether it was taken.

A task is an action bound to its arguments and to the callback it must notify - unlike [add_action\(\)](#), which queues work the caller hears nothing more about. It runs on whichever thread drives this execution, and the callback is invoked there too, with what the action returned. Queueing before [run\(\)](#) or [start\(\)](#) is normal.

Attention

The last argument is the callback, and this signature cannot say so: a parameter pack cannot be followed by a deducible parameter, so it arrives inside args and [untangle::bind_task\(\)](#) splits it off. It must be callable with the action's result and return nothing - or callable with nothing at all, when the action returns void, because finished is the message and the result is optional. A missing or unusable one is a `static_assert` rather than silence.

Finished does not mean failed. A task whose action throws is not notified - there is no result to hand over - and what it threw goes to [on_error](#), as a failing action's does. A caller waiting on the callback alone would wait for ever.

Remarks

The actions convention on [untangle::actuator](#) does not apply here and never did: it reads a trailing argument of an invocation, and a queued call has no invocation left to read from. That is what a task is for.

Template Parameters

Args	- The argument types to bind, of which the last is the callback.
------	--

Parameters

action	- The action to run, copied into the task.
args	- The arguments to bind to action, followed by the callback to notify.

Returns

true - queued, and this execution's worker will run it and notify the callback.

false - refused and dropped, so it will never run and never notify. Either the execution is stopped, or the task could not do its job: no callback to notify, or nothing to run. The answer is the whole

report - nothing is thrown and the callback is not invoked.

8.2.3.12 attach()

```
template<typename actionT>
template<typename otherT>
void untangle::async::execution< actionT >::attach (
    const std::shared_ptr< otherT > & other) [inline]
```

Attaches another execution, so that this one triggers it.

The attached execution's pending actions are run by this one's worker, and stopping this one stops that one too. Actions of different types can be run on one thread this way.

other is held weakly: a pass locks it for the call, so it is never destroyed while this execution's worker is running it, and one that is gone by the next pass is dropped there.

Parameters

other	- A std::shared_ptr owning the execution to trigger; requiring one keeps an unowned execution from being attached at all.
-------	---

Exceptions

invalid_attachment	- if other is attached already. An execution has at most one attacher, which keeps the graph a forest and the cycle check a walk rather than a search.
invalid_attachment	- if the attachment would close a cycle, itself included. Driving a cycle recurses until the stack is gone, so it is refused where the caller can still be told.

8.2.3.13 detach()

```
template<typename actionT>
template<typename otherT>
bool untangle::async::execution< actionT >::detach (
    otherT & other) [inline]
```

Detaches an execution previously attached by [attach\(\)](#).

Parameters

other	- The execution to stop triggering. Detaching one that was never attached is not an error: the caller asked for a state that already holds.
-------	---

Remarks

Refused, and nothing changed, from inside this execution's own attachment pass - from an action of other, say: its actuators are dispatching, and refuse removals then. Detach once the pass is over.

Returns

true - other was attached and is no longer.

false - other was not attached to this execution, or the detach was refused.

8.2.3.14 results()

```
template<typename actionT>
auto untangle::async::execution< actionT >::results () const [inline]
```

The return values of the actions run by [run\(\)](#), in the order they ran.

Read it from [on_finished](#), which fires once the queue has drained, so the results are complete by the time the callback sees them. Returned by value: the worker is appending to it.

Remarks

Only [run\(\)](#) fills this. A continuous worker has no point at which a run is over, so [start\(\)](#) collects nothing.

Returns

The results of the most recent run, or empty if none has produced any.

8.2.4 Member Data Documentation

8.2.4.1 action_execute

```
template<typename actionT>
std::function<void(void)> untangle::async::execution< actionT >::action_execute
```

Connection point that runs this execution's queued actions. Bound to [execute_actions\(\)](#).

Attention

Wired by the constructor. Assigning to it unwires whatever is connected, silently.

8.2.4.2 `action_stop`

```
template<typename actionT>
std::function<void(void)> untangle::async::execution< actionT >::action_stop
```

Connection point that stops this execution. Bound to [stop\(\)](#).

Attention

Wired by the constructor. Assigning to it unwires whatever is connected, silently.

8.2.4.3 `action_is_running`

```
template<typename actionT>
std::function<bool(void)> untangle::async::execution< actionT >::action_is_running
```

Connection point that reports whether this execution is running. Bound to [is_running\(\)](#).

[execution_poll](#) holds its address, and [~execution\(\)](#) takes it back out.

Attention

Wired by the constructor. Assigning to it unwires whatever is connected, silently.

8.2.4.4 `on_finished`

```
template<typename actionT>
std::function<void(void)> untangle::async::execution< actionT >::on_finished
```

Called on the worker's thread once the queue has drained. Assigned by the caller.

[results\(\)](#) is complete by the time it runs, and [is_running\(\)](#) reads true there.

Remarks

It is raised once per pass, not once per piece of work, and a pass that ran only tasks raises it like any other. Do not read it as "my task finished": that is the task's own callback, and it has already run by this point, inside the same drain and on this thread.

A worker that woke with an empty queue has finished nothing and stays silent.

8.2.4.5 on_error

```
template<typename actionT>
std::function<void(std::exception_ptr)> untangle::async::execution< actionT >::on_error
```

Called on the worker's thread with whatever an action threw. Assigned by the caller.

The action is dropped either way; this is what it is dropped to. Without it the only record is a warning on stderr naming this execution, which reaches a log and no code — an action that failed and one that succeeded are the same from outside. A handler assigned here replaces that warning, so a caller that takes responsibility for reporting is not reported twice.

Remarks

It receives a `std::exception_ptr` because the worker catches everything, including what is not a `std::exception`. Rethrow it to read it.

It is told about a dead binding too, which arrives as `untangle::invalid_action`. A caller asking why an action did not run wants both answers, not the one the header happens to raise itself.

A task's failure arrives here as well, and so does its callback's. A task whose action throws is not notified - there is no result to hand over - so this is the only report it makes. And because a callback runs inside the same try as the task that owns it, this can fire for a task whose action in fact succeeded: the work was done, and only the telling failed.

Attention

It runs on the worker's thread, inside the drain, like `on_finished`. Nothing may escape it: it is called from the same try the action was called from, and an exception leaving a detached thread function calls `std::terminate`.

The documentation for this class was generated from the following file:

- `async.hpp`

8.3 untangle::async::execution_poll Class Reference

Poll that reports whether any `execution` added to it is still running.

```
#include <async.hpp>
```

Public Member Functions

- `execution_poll ()=default`
Constructs an empty poll. `add()` is what puts anything in it.
- `~execution_poll ()`
Destroys the poll, releasing every execution still in it.
- `template<typename asyncexecT>`
`void add (asyncexecT &async_exec)`
Adds an `execution` object to the poll.
- `template<typename asyncexecT>`
`void remove (asyncexecT &async_exec)`
Removes an `execution` object from the poll.
- `void withdraw (const std::function< bool(void)> *action)`
Takes one action pointer out, without touching the execution it came from.
- `auto is_running ()`
Checks if the execution objects are running.

8.3.1 Detailed Description

Poll that reports whether any [execution](#) added to it is still running.

Remarks

An execution runs in a detached thread and cannot be joined, so checking the "running" state is the only way to wait for one. This checks any number of them at once.

A poll is constructed and owned by the caller, and answers for the executions added to that one. An owner of a set of workers - a thread pool is the case this was written for - waits for its own and not for whatever else the process is running.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 ~execution_poll()

```
untangle::async::execution_poll::~execution_poll () [inline]
```

Destroys the poll, releasing every execution still in it.

Remarks

The record runs both ways, and it has to: each side holds a pointer into the other, so whichever dies first takes itself out of the other. ~execution() withdraws from every poll holding it; this tells every execution it holds that this poll is gone. Neither order leaves a dangling pointer, which is what lets a poll be a local.

8.3.3 Member Function Documentation

8.3.3.1 add()

```
template<typename asyncexecT>
void untangle::async::execution_poll::add (
    asyncexecT & async_exec) [inline]
```

Adds an [execution](#) object to the poll.

Parameters

async_exec	An execution object.
------------	--------------------------------------

8.3.3.2 remove()

```
template<typename asyncexecT>
void untangle::async::execution_poll::remove (
    asyncexecT & async_exec) [inline]
```

Removes an [execution](#) object from the poll.

Attention

Parameters

async_exec	An execution object.
------------	--------------------------------------

8.3.3.3 withdraw()

```
void untangle::async::execution_poll::withdraw (
    const std::function< bool(void)> * action) [inline]
```

Takes one action pointer out, without touching the execution it came from.

What `~execution()` calls for each poll holding it: by then the object is being destroyed and must not be called back into, so the two halves of [remove\(\)](#) are separated here.

Parameters

action	- The address add() stored.
--------	---

8.3.3.4 is_running()

```
auto untangle::async::execution_poll::is_running () [inline]
```

Checks if the execution objects are running.

Returns

true - if at least one [execution](#) object in this poll is running.

The documentation for this class was generated from the following file:

- `async.hpp`

8.4 untangle::async::invalid_attachment Struct Reference

Raised by [execution::attach\(\)](#) when the attachment asked for is refused.

```
#include <async.hpp>
```

Public Member Functions

- [invalid_attachment](#) (std::string text)
Construct a new invalid attachment object.
- const char * [what](#) () const noexcept override
The message text describing the reason of this exception.

8.4.1 Detailed Description

Raised by `execution::attach()` when the attachment asked for is refused.

Remarks

Not an `untangle::invalid_action`: that one reports a binding whose target has died and is swallowed by the worker, while this is a caller error and must not be.

8.4.2 Constructor & Destructor Documentation

8.4.2.1 `invalid_attachment()`

```
untangle::async::invalid_attachment::invalid_attachment (
    std::string text) [inline], [explicit]
```

Construct a new invalid attachment object.

Parameters

text	- A message text, describing the reason of this exception.
------	--

8.4.3 Member Function Documentation

8.4.3.1 `what()`

```
const char * untangle::async::invalid_attachment::what () const [inline], [override], [noexcept]
```

The message text describing the reason of this exception.

Returns

`const char*` - The message text.

The documentation for this struct was generated from the following file:

- `async.hpp`

Chapter 9

File Documentation

9.1 async.hpp

```
00001 // Copyright (c) 2018 Nicolae Popescu. MIT License.
00002
00006 #ifndef UNTANGLE_ASYNC_ASYNC_HPP
00007 #define UNTANGLE_ASYNC_ASYNC_HPP
00008
00009 #include <actuator/actuator.hpp>
00010 #include <algorithm>
00011 #include <atomic>
00012 #include <chrono>
00013 #include <condition_variable>
00014 #include <cstdint>
00015 #include <exception>
00016 #include <functional>
00017 #include <iterator>
00018 #include <memory>
00019 #include <mutex>
00020 #include <print>
00021 #include <string>
00022 #include <thread>
00023 #include <type_traits>
00024 #include <utility>
00025 #include <vector>
00026
00027 #ifdef _MSC_VER
00028 #include <intrin.h>
00029 #endif
00030
00031 namespace untangle {
00032 namespace async {
00033
00035 inline void cpu_pause() noexcept {
00036 #if defined(_MSC_VER) && (defined(_M_X64) || defined(_M_IX86))
00037     __mm_pause();
00038 #elif defined(_MSC_VER) && defined(_M_ARM64)
00039     __yield();
00040 #elif defined(__x86_64__) || defined(__i386__)
00041     __builtin_ia32_pause();
00042 #elif defined(__aarch64__) || defined(__arm__)
00043     __asm__ __volatile__("yield");
00044 #endif
00045 }
00046
00055 inline std::unique_lock<std::mutex> lock_spinning(std::mutex& mutex) {
00056     constexpr int tries = 100;
00057     for (int attempt = 0; attempt < tries; ++attempt) {
00058         if (mutex.try_lock()) {
00059             return std::unique_lock<std::mutex>(mutex, std::adopt_lock);
00060         }
00061         cpu_pause();
00062     }
00063     return std::unique_lock<std::mutex>(mutex);
00064 }
00068
00075 struct invalid_attachment : std::exception {
00081     explicit invalid_attachment(std::string text) : message_(std::move(text)) {}
00082
00088     const char* what() const noexcept override { return message_.c_str(); }
```

```

00089
00090 private:
00091     std::string message_;
00092 };
00093
00102 struct attachment {
00104     std::weak_ptr<attachment> attacher;
00105 };
00106
00117 class execution_poll {
00118 public:
00122     execution_poll() = default;
00123
00132 ~execution_poll() {
00133     std::vector<registration> held;
00134
00135     {
00136         std::lock_guard<std::mutex> lock(actuator_mutex_);
00137         held.swap(registrations_);
00138     }
00139
00140     // Outside the lock: each of these takes the execution's own, and ~execution() takes them in
00141     // the opposite order. Releasing this one first is what keeps the two from meeting in a cycle.
00142     for (auto& one : held) {
00143         one.forget();
00144     }
00145 }
00146
00152 template <typename asyncexecT>
00153 void add(asyncexecT& async_exec) {
00154     {
00155         std::lock_guard<std::mutex> lock(actuator_mutex_);
00156
00157         if (!actuator_is_running_.is_connected()) {
00158             actuator_is_running_ = untangle::connect(async_exec.action_is_running);
00159         } else {
00160             actuator_is_running_.add(&async_exec.action_is_running);
00161         }
00162
00163         registrations_.emplace_back(&async_exec.action_is_running,
00164                                     [this, &async_exec] { async_exec.forget_poll(*this); });
00165     }
00166
00167     // Outside the lock, and deliberately: this takes the execution's own. The execution has to
00168     // know which polls hold a pointer into it, because ~execution() is what takes it back out and
00169     // there is now more than one poll it could be in.
00170     async_exec.remember_poll(*this);
00171 }
00172
00181 template <typename asyncexecT>
00182 void remove(asyncexecT& async_exec) {
00183     withdraw(&async_exec.action_is_running);
00184     async_exec.forget_poll(*this);
00185 }
00186
00195 void withdraw(const std::function<bool(void)>* action) {
00196     std::lock_guard<std::mutex> lock(actuator_mutex_);
00197
00198     actuator_is_running_.remove(action);
00199     std::erase_if(registrations_,
00200                 [action](const registration& one) { return one.action == action; });
00201 }
00202
00208 auto is_running() {
00209     std::lock_guard<std::mutex> lock(actuator_mutex_);
00210     actuator_is_running_();
00211
00212     auto result = false;
00213     for (const auto& ret : actuator_is_running_.results) {
00214         result = result || ret;
00215     }
00216     return result;
00217 }
00218
00219 private:
00224 struct registration {
00225     const std::function<bool(void)>* action;
00226     std::function<void()> forget;
00227 };
00228
00229 actuator<std::function<bool(void)>> actuator_is_running_;
00230 std::vector<registration> registrations_;
00231 mutable std::mutex actuator_mutex_;
00232 };
00233
00253 template <typename actionT>
00254 class execution {

```



```

00255 // An execution may attach one of a different specialisation, which is a different class;
00256 // attach() and detach() reach into its attachment_lifetime_.
00257 template <typename otherActionT>
00258 friend class execution;
00259
00260 // add() and remove() keep the record this object holds of the polls that hold it.
00261 friend class execution_poll;
00262
00263 public:
00273 template <typename... Args>
00274 static std::shared_ptr<execution> create_instance(Args&&... args) {
00275     return std::make_shared<execution>(std::forward<Args>(args)...);
00276 }
00277
00283 execution() : execution(std::string(default_name)) {}
00284
00290 explicit execution(std::string exec_name) : name(std::move(exec_name)) {
00291     action_execute = untangle::bind(this, &execution<actionT>::execute_actions);
00292     action_stop = untangle::bind(this, &execution<actionT>::stop);
00293     action_is_running = untangle::bind(this, &execution<actionT>::is_running);
00294 }
00295
00310 ~execution() {
00311     {
00312         auto lock = lock_spinning(action_mutex_);
00313         started_ = false;
00314         stopped_ = true;
00315     }
00316
00317     action_cv_.notify_all();
00318
00319     while (running_.load()) { // time of check
00320         std::this_thread::sleep_for(std::chrono::milliseconds(1));
00321     }
00322
00323     // time of use: this object is freed once the destructor returns, so the check above is only
00324     // safe because running_ is the last thing the worker touches.
00325     //
00326     // Out of every poll that holds a pointer into this object: an execution can be in several,
00327     // and an address left in any of them outlives the object. Taken by swap so nothing else can be
00328     // added to the record while this walks it, and withdrawn rather than removed because remove()
00329     // would call back into an object that is going away.
00330     std::vector<execution_poll*> holding_polls;
00331
00332     {
00333         std::lock_guard<std::mutex> lock(polls_mutex_);
00334         holding_polls.swap(polls_);
00335     }
00336
00337     for (auto* poll : holding_polls) {
00338         poll->withdraw(&action_is_running);
00339     }
00340 }
00341
00359 template <typename classT, typename T>
00360 static void bind_action_and_method(actionT& action, const std::shared_ptr<classT>& obj,
00361                                   T classT::* method,
00362                                   const std::shared_ptr<execution>& async_exec) {
00363     auto async_action = std::make_shared<const actionT>(untangle::bind(obj, method));
00364
00365     action = [wp = std::weak_ptr<execution>(async_exec),
00366              async_action](auto&&... args) -> actionT::result_type {
00367         // lock() also keeps the execution alive for the duration of the call
00368         const auto exec = wp.lock();
00369         if (!exec) {
00370             throw invalid_action("bind: invalid execution");
00371         }
00372
00373         exec->add_queued_action([async_action, ... args = std::forward<decltype(args)>(args)] {
00374             return (*async_action)(args...);
00375         });
00376
00377         return typename actionT::result_type();
00378     };
00379 }
00380
00392 template <typename T>
00393 static void bind_action_and_function(actionT& action, T Fn,
00394                                     const std::shared_ptr<execution>& async_exec) {
00395     auto async_action = std::make_shared<const actionT>(std::move(Fn));
00396
00397     action = [wp = std::weak_ptr<execution>(async_exec),
00398              async_action](auto&&... args) -> actionT::result_type {
00399         const auto exec = wp.lock();
00400         if (!exec) {
00401             throw invalid_action("bind: invalid execution");
00402         }

```

```

00403
00404     exec->add_queued_action([async_action, ... args = std::forward<decltype(args)>(args)] {
00405         return (*async_action)(args...);
00406     });
00407
00408     return typename actionT::result_type();
00409 };
00410 }
00411
00423 bool is_running() const { return running_.load(); }
00424
00440 bool is_busy() const {
00441     auto lock = lock_spinning(action_mutex_);
00442     return has_pending_actions_or_tasks() || executing_action_.load();
00443 }
00444
00453 std::size_t pending() const {
00454     auto lock = lock_spinning(action_mutex_);
00455     return action_actuator_.actions.size() + action_actuator_.actions_map.size() +
00456         action_actuator_.tasks.size();
00457 }
00458
00472 void run() {
00473     wait_thread_to_finish();
00474     collecting_results_ = true;
00475     thread_ = std::thread(&execution::execute, this);
00476     thread_.detach();
00477 }
00478
00489 void start() {
00490     wait_thread_to_finish();
00491     collecting_results_ = false;
00492     {
00493         auto lock = lock_spinning(action_mutex_);
00494         started_ = true;
00495         stopped_ = false;
00496     }
00497
00498     thread_ = std::thread(&execution::loop, this);
00499     thread_.detach();
00500 }
00501
00508 void stop() {
00509     {
00510         auto lock = lock_spinning(action_mutex_);
00511         started_ = false;
00512         stopped_ = true;
00513     }
00514
00515     // The worker drains what is already queued before it leaves loop(), so stop() does not have to
00516     // spin on the queue; waking it is enough.
00517     action_cv_.notify_all();
00518
00519     if (actuator_stop_.is_connected()) {
00520         actuator_stop_();
00521     }
00522 }
00523
00551 template <typename... Args>
00552 bool add_action(actionT action, Args&&... args) {
00553     return add_queued_action(std::bind(std::move(action), std::forward<Args>(args)...));
00554 }
00555
00588 template <typename... Args>
00589 bool add_task(const actionT& action, Args&&... args) {
00590     return add_queued_task(untangle::bind_task(action, std::forward<Args>(args)...));
00591 }
00592
00597 template <typename... Args>
00598 bool add_task(actionT&& action, Args&&... args) {
00599     return add_queued_task(untangle::bind_task(std::move(action), std::forward<Args>(args)...));
00600 }
00601
00620 template <typename otherT>
00621 void attach(const std::shared_ptr<otherT>& other) {
00622     if (static_cast<const void*>(other.get()) == static_cast<const void*>(this)) {
00623         throw invalid_attachment("attach: an execution cannot be attached to itself");
00624     }
00625
00626     if (!other->attachment_lifetime_->attacher.expired()) {
00627         throw invalid_attachment("attach: that execution is attached already");
00628     }
00629
00630     // At most one attacher each makes the graph a forest, so a cycle closes exactly when other is
00631     // somewhere up this chain. Walking it is the whole check.
00632     for (auto link = attachment_lifetime_->attacher.lock(); link; link = link->attacher.lock()) {
00633         if (link == other->attachment_lifetime_) {

```

```

00634         throw invalid_attachment("attach: the attachment would close a cycle");
00635     }
00636 }
00637
00638 // Bound weakly, as owned actions: a call keeps other alive, a call to one that is gone throws
00639 // untangle::invalid_action, and the actuator drops it.
00640 std::erase_if(attachments_, [](const auto& entry) { return entry.attached.expired(); });
00641 attachments_.push_back({other->attachment_lifetime_,
00642     actuator_execute_.add(untangle::bind(other, &otherT::execute_actions)),
00643     actuator_stop_.add(untangle::bind(other, &otherT::stop))});
00644 other->attachment_lifetime_->attacher = attachment_lifetime_;
00645
00646 // An idle worker blocks until it is told otherwise; from now on it has to poll.
00647 {
00648     auto lock = lock_spinning(action_mutex_);
00649     attached_ = true;
00650 }
00651 action_cv_.notify_all();
00652 }
00653
00654 template <typename otherT>
00655 bool detach(otherT& other) {
00656     if (actuator_execute_.is_dispatching() || actuator_stop_.is_dispatching()) {
00657         return false;
00658     }
00659
00660     const auto entry = std::ranges::find_if(attachments_, [&other](const auto& candidate) {
00661         return candidate.attached.lock() == other.attachment_lifetime_;
00662     });
00663     if (entry == attachments_.end()) {
00664         return false;
00665     }
00666
00667     actuator_execute_.remove(entry->execute);
00668     actuator_stop_.remove(entry->stop);
00669     attachments_.erase(entry);
00670
00671     other.attachment_lifetime_->attacher.reset();
00672
00673     {
00674         auto lock = lock_spinning(action_mutex_);
00675         attached_ = actuator_execute_.is_connected();
00676     }
00677
00678     return true;
00679 }
00680
00681 auto results() const {
00682     std::lock_guard<std::mutex> lock(results_mutex_);
00683     return results_;
00684 }
00685
00686 static constexpr auto default_name = "default_name";
00687
00688 std::function<void(void)> action_execute;
00689
00690 std::function<void(void)> action_stop;
00691
00692 std::function<bool(void)> action_is_running;
00693
00694 std::function<void(void)> on_finished;
00695
00696 std::function<void(std::exception_ptr)> on_error;
00697
00698 std::string name = default_name;
00699
00700 private:
00701     using queued_action_t = std::function<typename actionT::result_type(void)>;
00702
00703     void remember_poll(execution_poll& poll) {
00704         std::lock_guard<std::mutex> lock(polls_mutex_);
00705         polls_.push_back(&poll);
00706     }
00707
00708     void forget_poll(execution_poll& poll) {
00709         std::lock_guard<std::mutex> lock(polls_mutex_);
00710         std::erase(polls_, &poll);
00711     }
00712
00713     bool has_pending_actions_or_tasks() const {
00714         return action_actuator_.is_connected() || action_actuator_.has_tasks();
00715     }
00716
00717     bool add_queued_action(queued_action_t action) {
00718         bool wake = false;

```

```

00826 {
00827     auto lock = lock_spinning(action_mutex_);
00828
00829     // Once stopped, the worker is on its way out and would never reach this action; dropping it
00830     // here is what keeps it from sitting in the queue looking as though it were pending.
00831     if (stopped_) {
00832         std::println(stderr, "warning: execution '{}{}' is stopped, action not added", name);
00833         return false;
00834     }
00835
00836     action_actuator_.add(std::move(action));
00837     wake = sleeping_;
00838 }
00839
00840 // A worker that has not parked finds the work on its own.
00841 if (wake) {
00842     action_cv_.notify_one();
00843 }
00844 return true;
00845 }
00846
00861 bool add_queued_task(untangle::task_t task) {
00862     bool wake = false;
00863     {
00864         auto lock = lock_spinning(action_mutex_);
00865
00866         // As for an action: a stopped worker would never reach it, and a task left in the queue
00867         // would look pending while being unable to ever notify.
00868         if (stopped_) {
00869             std::println(stderr, "warning: execution '{}{}' is stopped, task not added", name);
00870             return false;
00871         }
00872
00873         if (!action_actuator_.add_task(std::move(task))) {
00874             std::println(stderr, "warning: execution '{}{}' refused a task that cannot report", name);
00875             return false;
00876         }
00877         wake = sleeping_;
00878     }
00879
00880 // A worker that has not parked finds the work on its own.
00881 if (wake) {
00882     action_cv_.notify_one();
00883 }
00884 return true;
00885 }
00886
00896 void execute_actions() {
00897     actions_and_tasks_run_ = 0;
00898
00899     for (;;) {
00900         // Takes over action_actuator_ under the lock; its actions then run without it.
00901         actuator<queued_action_t> batch;
00902
00903         {
00904             auto lock = lock_spinning(action_mutex_);
00905             if (!has_pending_actions_or_tasks()) {
00906                 break;
00907             }
00908
00909             batch = std::move(action_actuator_);
00910
00911             // Under the same lock, so is_busy() sees both or neither.
00912             executing_action_ = true;
00913         }
00914
00915         // Counted before either call: a dead binding leaves the actions list during operator(),
00916         // and call_tasks() consumes the tasks list outright.
00917         const auto in_the_batch = batch.actions.size() + batch.tasks.size();
00918
00919         // The actions have their own arms inside the actuator; this one is for the work around
00920         // them, and for the rule that nothing may leave this loop: an exception escaping a detached
00921         // thread function calls std::terminate.
00922         try {
00923             batch();
00924
00925             // Empty for a void action type, which is what makes this one line rather than two arms.
00926             if (collecting_results_) {
00927                 std::lock_guard<std::mutex> lock(results_mutex_);
00928                 std::ranges::move(batch.results, std::back_inserter(results_));
00929             }
00930
00931             // After the actions, and deliberately: one pass fires every action it holds and then every
00932             // task, so within a batch the two kinds do not interleave in the order they were queued.
00933             // Across batches the queue is still first in, first out. A task reports to its own
00934             // callback, so nothing is collected here; what one throws lands in batch.errors below,
00935             // beside what an action threw.

```

```

00936     batch.call_tasks();
00937 } catch (...) {
00938     report_error(std::current_exception());
00939 }
00940
00941 for (const auto& thrown : batch.errors) {
00942     report_error(thrown);
00943 }
00944
00945 actions_and_tasks_run_ += in_the_batch;
00946
00947 executing_action_ = false;
00948
00949 // The notification is raised without the lock: on_finished may call add_action(), which
00950 // takes it.
00951 bool drained = false;
00952 {
00953     auto lock = lock_spinning(action_mutex_);
00954     drained = !has_pending_actions_or_tasks();
00955 }
00956
00957 if (drained) {
00958     notify_finished();
00959 }
00960 }
00961
00962 // Last, and part of the pass: an attachment is only ever reached through execute_actions(),
00963 // which is this function, so whoever drives this execution drives the ones attached to it.
00964 if (actuator_execute_.is_connected()) {
00965     actuator_execute_();
00966
00967     // The pass dropped the last attachment, gone since: the worker can block again.
00968     if (!actuator_execute_.is_connected()) {
00969         auto lock = lock_spinning(action_mutex_);
00970         attached_ = false;
00971     }
00972 }
00973 }
00974
00975 void report_error(std::exception_ptr thrown) {
00976     if (on_error) {
00977         // The handler is the caller's code, on a detached thread; anything escaping it would leave
00978         // the thread function and call std::terminate.
00979         try {
00980             on_error(thrown);
00981         } catch (...) {
00982             std::println(stderr,
00983                 "warning: execution '{}{}' had on_error throw, and dropped what it threw", name);
00984         }
00985     }
00986     return;
00987 }
00988
00989 try {
00990     std::rethrow_exception(thrown);
00991 } catch (const invalid_action& ia) {
00992     std::println(stderr, "warning: execution '{}{}' dropped an invalid action: {}", name,
00993         ia.what());
00994 } catch (const std::exception& e) {
00995     std::println(stderr, "warning: execution '{}{}' dropped an action that threw: {}", name,
00996         e.what());
00997 } catch (...) {
00998     std::println(stderr, "warning: execution '{}{}' dropped an action that threw an unknown type",
00999         name);
01000 }
01001 }
01002
01003 void notify_finished() {
01004     if (actions_and_tasks_run_ == 0 || !on_finished) {
01005         return;
01006     }
01007
01008     {
01009         // An action may queue another, so a pass that drained can leave more behind it. That is the
01010         // next batch, not the end of this one.
01011         auto lock = lock_spinning(action_mutex_);
01012         if (has_pending_actions_or_tasks()) {
01013             return;
01014         }
01015     }
01016
01017     on_finished();
01018 }
01019
01020 void wait_thread_to_finish() {
01021     auto resident = false;
01022     while (!running_.compare_exchange_weak(resident, true)) {

```

```

01042     resident = false;
01043     std::this_thread::sleep_for(std::chrono::milliseconds(1));
01044 }
01045 }
01046
01047 void execute() {
01048     // The results belong to this run, not to this run and the last one.
01049     {
01050         std::lock_guard<std::mutex> lock(results_mutex_);
01051         results_.clear();
01052     }
01053
01054     execute_actions();
01055
01056     // Must stay last: ~execution() may free this object the moment it reads false.
01057     running_ = false;
01058 }
01059
01060 void loop() {
01061     for (;;) {
01062         {
01063             auto lock = lock_spinning(action_mutex_);
01064
01065             const auto has_work = [this] { return has_pending_actions_or_tasks() || !started_; };
01066
01067             // Polls only while something is attached: an attached execution has its own list and
01068             // cannot notify this condition variable. With nothing attached the worker blocks until it
01069             // has work, is stopped, or attach() wakes it.
01070             // Parked in either wait, and under the lock: an add reads it under the same lock, so it
01071             // notifies exactly when the worker may be waiting.
01072             sleeping_ = true;
01073             if (attached_) {
01074                 action_cv_.wait_for(lock, std::chrono::milliseconds(10), has_work);
01075             } else {
01076                 action_cv_.wait(lock, [this, &has_work] { return has_work() || attached_; });
01077             }
01078             sleeping_ = false;
01079
01080             if (!started_ && !has_pending_actions_or_tasks()) {
01081                 break;
01082             }
01083         }
01084         execute_actions();
01085     }
01086
01087     // A last pass for the attachments; this execution's own list is already empty here.
01088     execute_actions();
01089
01090     // Must stay last: ~execution() may free this object the moment it reads false.
01091     running_ = false;
01092 }
01093
01094 std::thread thread_;
01095
01096 std::atomic_bool started_ = {false};
01097 std::atomic_bool running_ = {false};
01098
01099 // Set by stop(), and the difference between "not started yet" and "finished for good": actions
01100 // may be queued before run()/start(), but not after stop().
01101 std::atomic_bool stopped_ = {false};
01102
01103 // action_actuator_, started_ and stopped_ are written by every thread that calls add_action() or
01104 // stop() and read by the worker; nothing touches them outside this mutex.
01105 mutable std::mutex action_mutex_;
01106 std::condition_variable action_cv_;
01107
01108 bool sleeping_ = false;
01109
01110 bool attached_ = false;
01111
01112 actuator<queued_action_t> action_actuator_;
01113
01114 using void_actuator = actuator<std::function<void(void)>>;
01115
01116 void_actuator actuator_execute_;
01117 void_actuator actuator_stop_;
01118
01119 std::shared_ptr<attachment> attachment_lifetime_ = std::make_shared<attachment>();
01120
01121 struct attached_entry {
01122     std::weak_ptr<attachment> attached;
01123     const std::function<void(void)>* execute = nullptr;
01124     const std::function<void(void)>* stop = nullptr;
01125 };
01126 std::vector<attached_entry> attachments_;
01127
01128 // std::vector cannot hold void type; use an arbitrary type e.g. int

```

```
01169 using resultT = std::conditional<std::is_void<typename actionT::result_type>::value, int,
01170                                     typename actionT::result_type>;
01171
01175 std::vector<typename resultT::type> results_;
01176
01180 mutable std::mutex results_mutex_;
01181
01188 std::atomic_bool collecting_results_ = {false};
01189
01195 std::atomic_bool executing_action_ = {false};
01196
01198 std::vector<execution_poll*> polls_;
01199 mutable std::mutex polls_mutex_;
01200
01212 std::atomic_size_t actions_and_tasks_run_ = {0};
01213 };
01214 } // namespace async
01215 } // namespace untangle
01216
01217 #endif // UNTANGLE_ASYNC_ASYNC_HPP
```


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